



Solar Project at Minnehaha United Methodist Church

Solar Committee - Chris Kramer, Joan Ellison, Mike Schmidt
Eric Strom, John Fairbairn, Beth Carlson

Project Sponsors - MUMC Trustees & Pastor Becky Sechrist

Minnehaha United Methodist Church Mission Statement

We seek, we serve, we celebrate,
we carry our Christian faith into our
community and the world. We
provide a caring gathering place to
worship God and to grow spiritually.

Why Consider Solar Power Now?

- MUMC community interested in reduced consumption of fossil fuels
- Conservation efforts undertaken
Energy audit (2015)
Additional efforts ongoing...AND
- A window of opportunity

Solar Committee Goals

- Evaluate paths to solar (onsite rooftop and/or community solar garden)
- Determine ROI and cost feasibility and best option for MUMC
- Implement solar option (if approved)

Project Planning

- ☒ Planning Phase - COMPLETED
- ☐ Decision / Approval - proposal is ready
- ☐ Execution / Project management

Solar Committee Findings

- On-site solar panel installation provides highest ROI and is viable
- Receiving state grant funding makes this the best time for this kind of endeavor

Benefits

- Reduced carbon footprint
~42% reduction
- Savings of \$250K-\$300K over 35 years
- New monitoring tools to better assess electricity usage
- Visible sign of commitment to environmental sustainability

How Is This Viable?

1. Made in Minnesota Incentive Program
(rebate from solar production)
2. Installation Partner - Apadana
3. MinnPACE Financing
4. Federal Tax Rebate of 15%
(via CollectiveSun)

1. Made in Minnesota (MiM)

- State incentive program
- Maximum installation size: 40 kW
- Ten-year program pays incentive rebate of \$0.14/kWh produced by the solar array
- MUST have system installed and running by October 31st, 2018!

2. Solar Installer & Electrical Contractor



<https://www.apadanasolartech.com>

2. Solar Installer & Electrical Contractor



- Solar PV Installer Since 2007
- Based in Golden Valley, MN
- Over 1750 projects
- Warranty - 10 years (Labor)
- 25 years (Materials)

3. Lender / Financing Partner



MinnPACE

<https://www.minnpace.com>

3. Property Assessed Clean Energy (PACE)



MinnPACE

- For renewable energy upgrades
- Other local sites (including churches) successfully used it
- 10-year term at 5% APR

4. Leasing Partner



<https://www.collectivesun.com>

4. About CollectiveSun



- Helps nonprofits fund solar projects, nationwide. CollectiveSun is a for-profit company able to take advantage of federal tax credits
- Applies tax credits that reduce the cost of non-profit solar projects by 15%

Questions



What is MUMC Electricity Usage?

- Monthly: 10,000 kilowatt hours (kWh)
- Annual: 120,000 kWh
- Annual: 96 KW
- Base rate from Xcel: \$0.09 per kWh
- Annual electricity bills: avg \$14,400 (2015-17)

NOTE: This is the total bill from Xcel and includes many other charges on the bill beyond usage cost. We calculate our blended rate at \$0.13 per kWh.

What Is Our Capacity for Rooftop Solar?

Viable areas for rooftop solar are

- Education building (30-40 kW)
- Sanctuary (15-20 kW)

Proposed project is 40kW system

- Aligns with MiM maximum
- 50% Sanctuary / 50% Education building

Apadana Project Details

- Workmanship, panels, inverters all warrantied 10-25 years
- Panel production warrantied for 25 years
- Includes online monitoring system
- CollectiveSun owns panels in years 1-6;
MUMC owns panels year 7+

How are payments made?

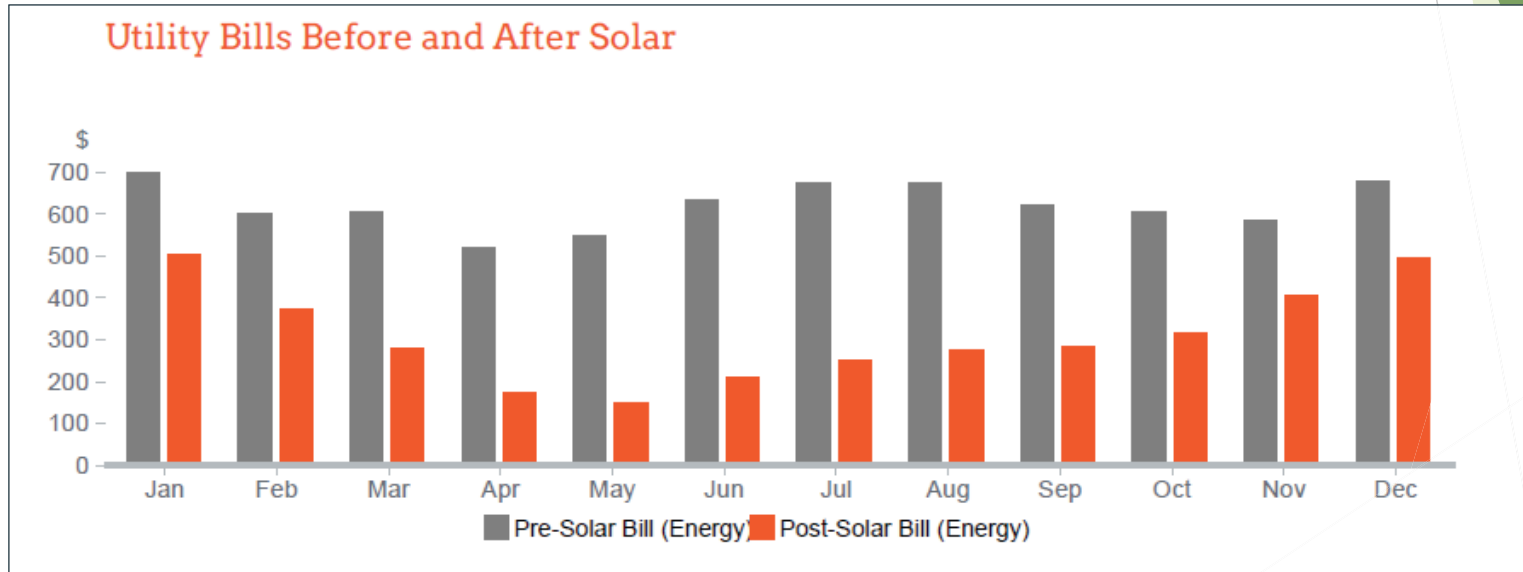


MinnPACE

- Payments rolled into special tax assessment due with regular tax payments
- 1st payment is due in May of the year following installation (May 2019)
- Finance would create fund to allocate money from electricity savings, MiM incentives for payments

What Would Be Our Energy Savings?

- Significantly reduce summer usage; less effect in winter



What Is The Overall Project Cost?

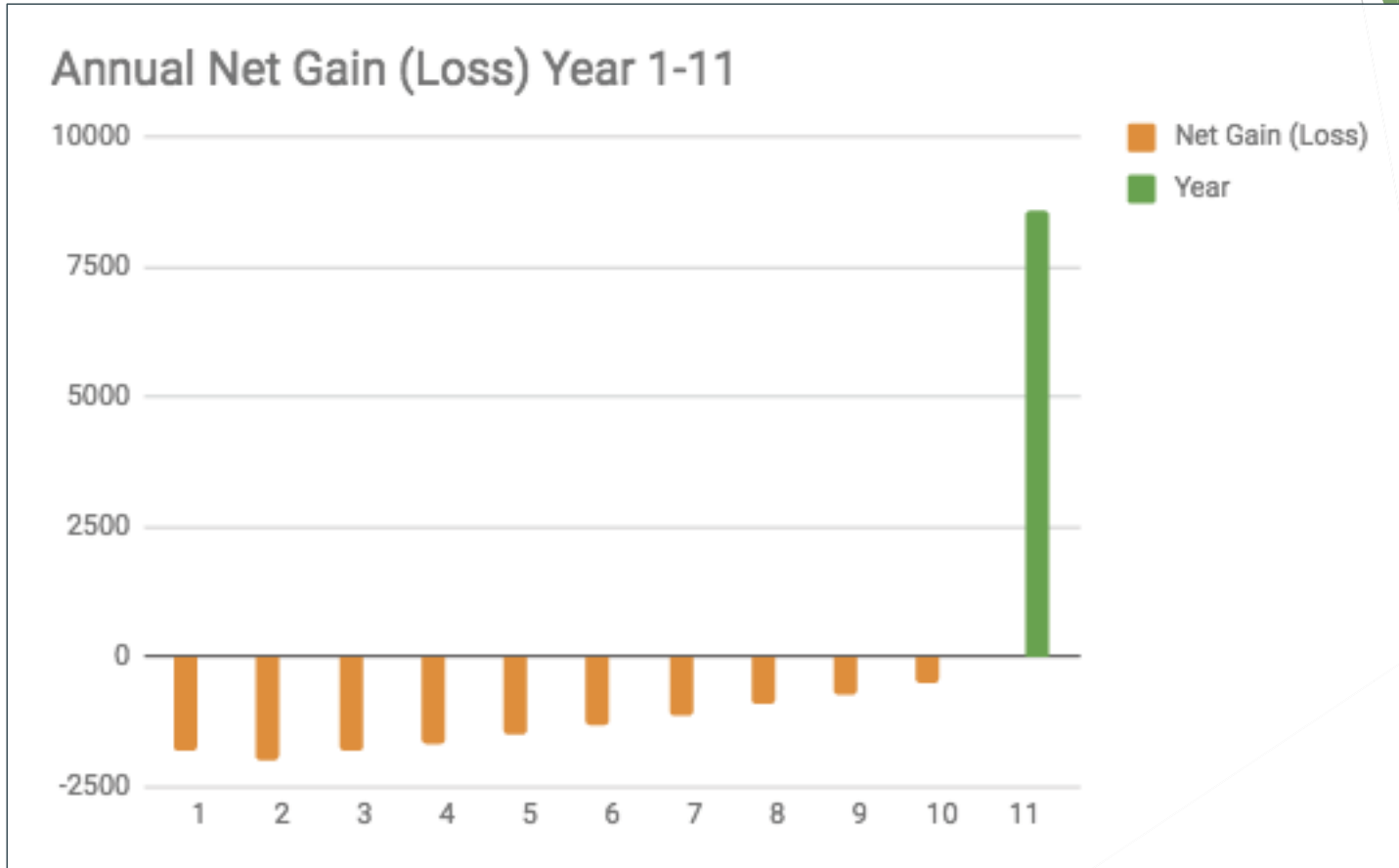
- Year 1 overview:

System Cost	\$136,896
Collective Sun 15% credit	\$20,534
Net Project Cost	\$116,362
Loan Amount Principal	\$121,500
Interest Total	\$34,378
P & I Total	\$155,878
Annual Payments	\$15,588
Annual Solar Savings	\$13,795
Difference to be paid (Year 1)	(\$1793)

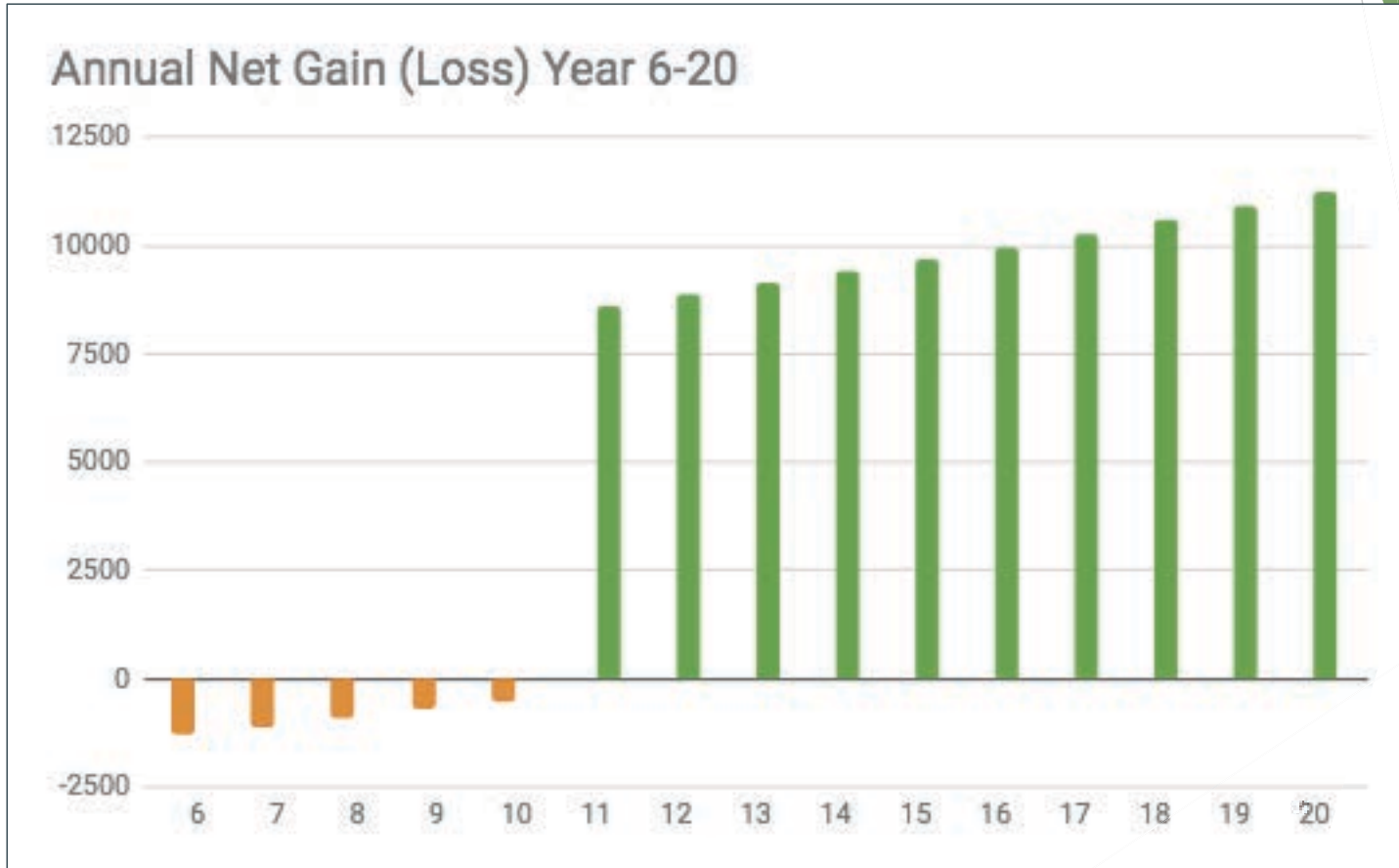
How Would Cash Flow Look?

Year	Annual Solar Savings	Annual Loan Payment	Cash Flow	Cumulative Cash Flow
1	13,795	15,588	(1,793)	(1,793)
2	13,605	15,588	(1,983)	(3,776)
3	13,767	15,588	(1,821)	(5,597)
4	13,936	15,588	(1,652)	(7,249)
5	14,110	15,588	(1,478)	(8,727)
6	14,291	15,588	(1,297)	(10,024)
7	14,479	15,588	(1,109)	(11,133)
8	14,673	15,588	(915)	(12,048)
9	14,875	15,588	(713)	(12,761)
10	15,083	15,588	(505)	(13,266)
11*	8,617	0	8,617	(4,648)

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...				
20	11,227	0	11,227	85,445
25	13,004	0	13,004	146,805
35	17,446	0	17,446	300,201

Potential Risks? - Condition of Roof

- Education building roof not in great shape, but also not leaking
- Roofer verified a ballast system for panels is best (no penetration of existing roof)
- System would be removed / reinstalled at the time Education bldg needs roof replaced (Cost would be rolled into any roof repair project)

Potential Risks? - Delayed Completion

- Hard deadline of getting system online by October 31, 2018 to receive MiM incentives (Est value: \$68,874)
- We want to get system installed and online ASAP to start production and benefits.
- Summer is best season for production

Potential Risks? - Low Solar Production

- Estimates of production from Apadana are conservative. Very likely to meet estimates.
- Insurance coverage for damage to array
- CollectiveSun responsible for repairs/replacement during lease period

Next Steps

- Review with congregation (happening now)
- Informational meeting
Wednesday, May 16th 6:30 pm
- Decision from congregation to proceed
Vote is Sunday, June 3rd at 11:15 am



Links of Interest

CollectiveSun (Lease Partner)

www.collectivesun.com

Made in MN (State Incentive)

<https://mn.gov/commerce/industries/energy/solar/mim/>

St. Paul Port Authority MinnPACE (Financing)

<https://www.sppa.com/energy-financing/minnpace>

Apadana (Contractor / Installer)

<https://www.apadanasolartech.com>